SentinelForge IR
Incident response & security engineering

Endpoint Connection Manual

How to connect Ubuntu Desktop, Ubuntu Server and Windows machines to SentinelForge IR so their sign-ins, admin activity and security events are collected and checked by the detections.

Ubuntu Desktop 20.04+ | Ubuntu Server 20.04+ | Windows 10 / 11 | Windows Server 2016+ | Sysmon (optional)

Application
https://sentinelforge.kuinkel.com

Ingestion API
https://sentinelforge-api.kuinkel.com

Version
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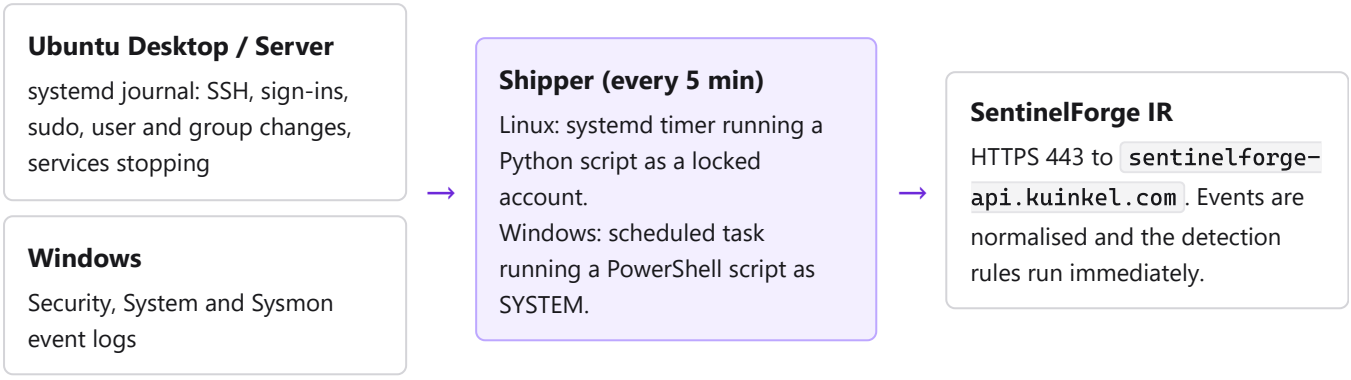
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About the pictures in this manual

Screenshots of the SentinelForge web app are real captures of the application, taken from a test workspace filled with synthetic data (documentation IP addresses such as 203.0.113.50 and invented host and user names). Terminal windows are drawn illustrations of the commands and the output the scripts print; your host names, counts and times will differ.

1. How endpoint collection works

A small shipper runs on each machine every five minutes. It reads the machine's own security logs since its last run and posts them over HTTPS to SentinelForge with that machine's ingest key. Nothing listens on the machine and no inbound port is opened.



What each machine contributes

Machine	Collected	Detections that use it
Ubuntu Desktop	Greeter (GDM/LightDM) and console sign-ins and failures, `sudo` commands and failed `sudo` / `su` passwords, account and group changes	SF-IAM-001 new privileged account; investigations
Ubuntu Server	SSH logins and failures with source IP, `sudo` commands, `useradd` / `usermod` / `gpasswd`, `auditd` / `rsyslog` / `journald` stopping	SF-AUTH-001 password guessing, SF-IAM-001, SF-DEF-001 logging stopped
Windows	Logons (with logon type), process creation, scheduled task creation, account and group changes, log clearing, services stopping; with Sysmon also LSASS access, Run-key writes, network and DNS	SF-AUTH-001, SF-PROC-001, SF-IAM-001, SF-DEF-001, SF-CRED-001, SF-PERS-001/002, SF-LAT-001

Before you start

- ☐ A SentinelForge **admin** account (demo accounts cannot create data sources).
- ☐ Administrator rights on each machine: `sudo` on Ubuntu, a local Administrator on Windows.
- ☐ Outbound HTTPS (TCP 443) from each machine to `sentinelforge-api.kuinkel.com`. Check with `curl -sI https://sentinelforge-api.kuinkel.com/api/v1/health` (Ubuntu) or `Invoke-RestMethod https://sentinelforge-api.kuinkel.com/api/v1/health` (Windows).
- ☐ The connector files from the repository: `connectors/linux/` (two files) and `connectors/windows/Send-WindowsEvents.ps1`.

> **Treat ingest keys like passwords**
> Anyone with a key can send events as that machine. Paste it only into the installer prompt or the machine-level setting described below, never into scripts, tickets or chat. If a key leaks, rotate it (chapter 7).

2. Create a data source and ingest key

Create one data source per machine. Each gets its own key, and Settings shows each machine's health separately.

1. Sign in at `https://sentinelforge.kuinkel.com` as an admin and open **Settings** (bottom of the left menu). Scroll to **Data sources** and click **Add source**.
2. Name it after the machine's host name and choose the type:

Machine	Type
Ubuntu Desktop or Server	`linux`
Windows (with or without Sysmon)	`windows_security`

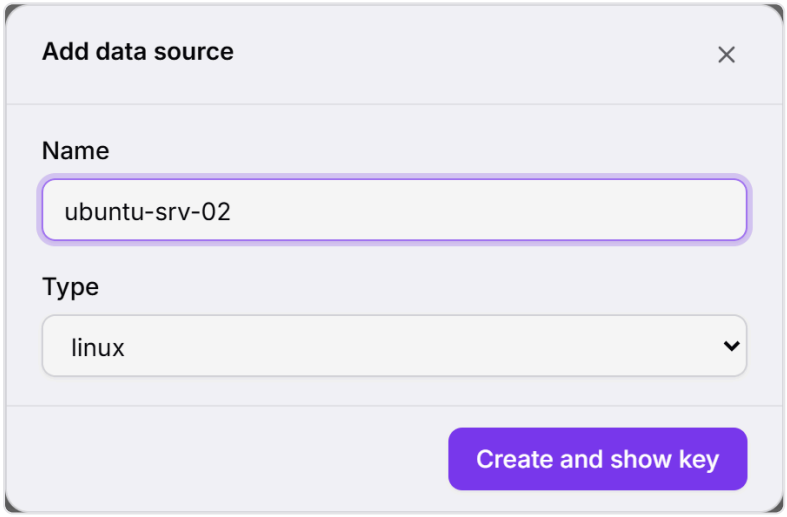


Figure 1. Adding a data source for an Ubuntu server.

3 Click **Create and show key**. Copy the key that starts with `sfk_` with the **Copy** button. It is shown **once**; only a hash is stored. Click **I have stored it**.

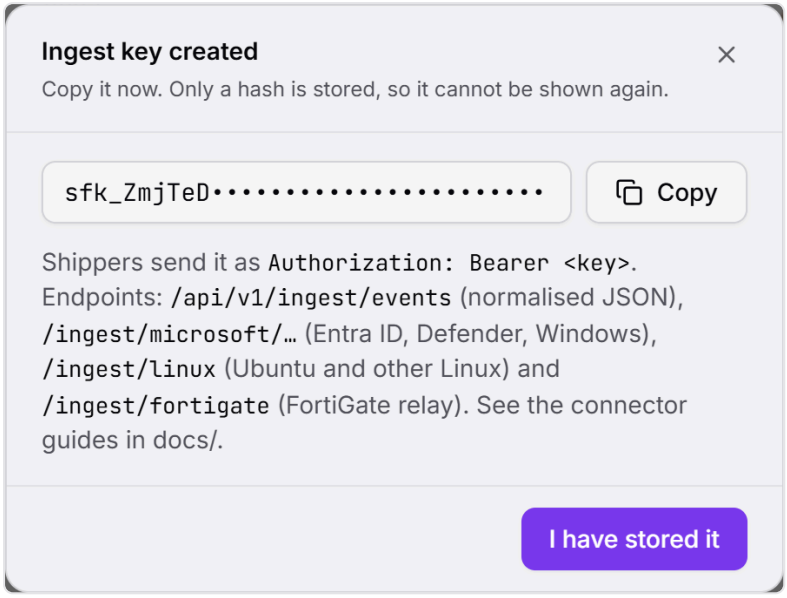


Figure 2. The one-time key dialog (key masked in this manual).

Once a machine sends its first events, its source shows **Reporting** with an event count and the time of the last event:

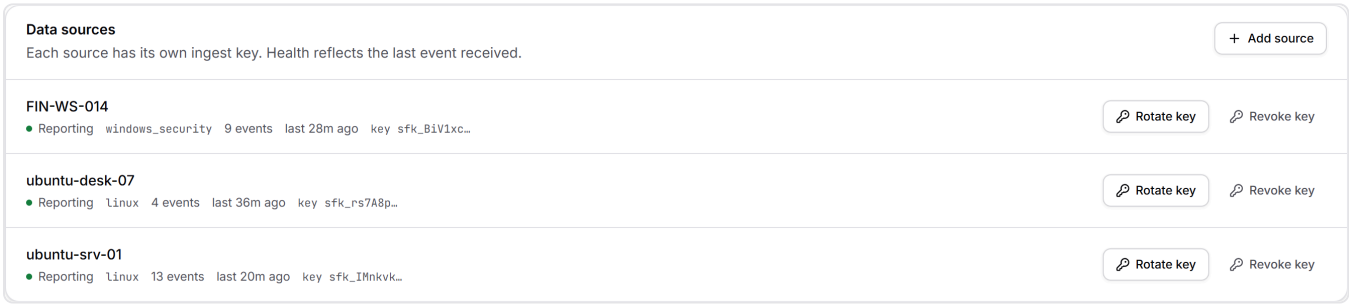


Figure 3. Three connected machines: a Windows workstation, an Ubuntu desktop and an Ubuntu server.

3. Ubuntu Desktop

Works on Ubuntu Desktop 20.04, 22.04, 24.04 and later (GNOME, and other desktops that use LightDM or SDDM). The installer sets everything up; you need the two files from `connectors/linux/` and the ingest key from chapter 2.

3.1 Get the files onto the desktop

Use whichever is easiest:

- **USB stick:** copy the `connectors\linux` folder from the repository to the stick, plug it into the Ubuntu desktop, and copy the folder to your home folder as `sentinelforge` using the Files app.
- **From a Windows PC over the network:** on the Ubuntu desktop install the SSH server once (`sudo apt install -y openssh-server`) and find its address with `hostname -I`. Then in PowerShell on the Windows PC:

```
scp -r C:\path\to\sentinal-IR\connectors\linux  maya@192.168.1.50:~/sentinelforge
```

3.2 Install

1. Open **Terminal**: press **Ctrl + Alt + T**, or open **Show Apps** and search for "Terminal".
2. Run the installer and paste the ingest key when asked. Nothing appears while you paste; that is normal for secret input. Press **Enter**.

```
maya@ubuntu-desk-07: ~/sentinelforge

maya@ubuntu-desk-07:~$ cd ~/sentinelforge
maya@ubuntu-desk-07:~/sentinelforge$ sudo bash install.sh --api-url https://sentinelforge-
api.kuinkel.com
[sudo] password for maya:
Ingest key for this host's data source (sfk_...):
Running a first collection...
2026-10-28T13:40:02Z read 1184 journal records, sent 9, 9 new events, 0 alerts raised
sentinelforge-shipper.service: Deactivated successfully.
Finished sentinelforge-shipper.service - Ship security events from the journal to
SentinelForge IR.
Installed. Events are sent every 5 minutes. Check: systemctl list-timers sentinelforge-
shipper.timer
```

Illustration of the expected output. "sent 9" means 9 of the journal records were security events SentinelForge uses.

3.3 What the installer set up

Item	Purpose
`sentinelforge` system account	Runs the shipper. No password, no shell, no home, no sudo; member of `systemd-journal` only, so it can read logs and nothing else.
`/opt/sentinelforge/`	The shipper script.

`/etc/sentinelforge/linux-shipper.env`	API URL and ingest key, readable only by root and the shipper account.
`sentinelforge-shipper.service` and `.timer`	Runs every 5 minutes (and 2 minutes after boot), with systemd hardening.
`/var/lib/sentinelforge/journal.cursor`	Where it got to in the log, so nothing is sent twice or skipped.

3.4 Check it on the desktop

Lock the screen (**Super + L**), type a wrong password once, then sign in properly. After the next run (up to 5 minutes) both appear in SentinelForge. Locally:

```
maya@ubuntu-desk-07: ~

maya@ubuntu-desk-07:~$ systemctl list-timers sentinelforge-shipper.timer
NEXT                        LEFT     LAST                        PASSED  UNIT
ACTIVATES
Tue 2026-10-28 13:50:31 UTC 3min 4s  Tue 2026-10-28 13:45:31 UTC 1min ago sentinelforge-
shipper.timer sentinelforge-shipper.service
maya@ubuntu-desk-07:~$ journalctl -u sentinelforge-shipper.service -n 1 -o cat --grep read
2026-10-28T13:45:33Z read 214 journal records, sent 2, 2 new events, 0 alerts raised
```

Illustration of the expected output.

> **Desktop sign-ins have no source IP**
> Because they happen at the keyboard, they are recorded for investigations but do not trigger the password-guessing rule, which needs a remote address.

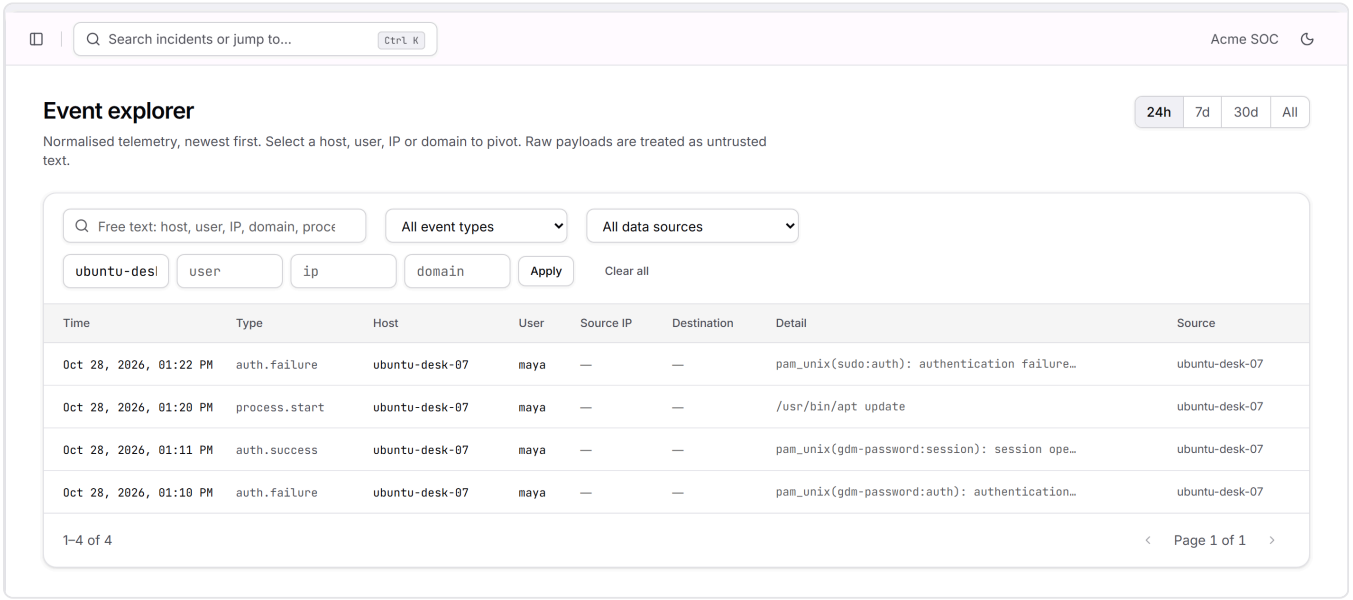


Figure 4. Events → host `ubuntu-desk-07`: a failed and a successful desktop sign-in, a `sudo apt update`, and a failed `sudo` password.

4. Ubuntu Server

Works on Ubuntu Server 20.04 and later, including cloud images. The steps are the same as for the desktop, done over SSH.

4.1 Copy the files and connect

From the Windows PC that has the repository (PowerShell), copy the folder, then sign in to the server:

```
scp -r C:\path\to\sentinal-IR\connectors\linux  deploy@203.0.113.10:~/sentinelforge
ssh deploy@203.0.113.10
```

Using PuTTY or WinSCP instead works the same way: copy the two files into `~/sentinelforge` and open a session.

4.2 Make sure the journal is kept on disk

Standard installs keep it in `/var/log/journal`. Minimal and some cloud images keep it only in memory, so events from before a reboot would be lost:

```
ls -d /var/log/journal || { sudo mkdir -p /var/log/journal && sudo systemctl restart systemd-journald; }
```

4.3 Install

deploy@ubuntu-srv-01: ~/sentinelforge

```
deploy@ubuntu-srv-01:~$ cd ~/sentinelforge && ls
install.sh  sentinelforge_linux_shipper.py
deploy@ubuntu-srv-01:~/sentinelforge$ sudo bash install.sh --api-url https://sentinelforge-api.kuinkel.com
Ingest key for this host's data source (sfk_...):
Running a first collection...
2026-10-28T13:41:10Z read 3907 journal records, sent 13, 13 new events, 3 alerts raised
sentinelforge-shipper.service: Deactivated successfully.
Finished sentinelforge-shipper.service - Ship security events from the journal to SentinelForge IR.
Installed. Events are sent every 5 minutes. Check: systemctl list-timers sentinelforge-shipper.timer
```

Illustration of the expected output.

To see exactly what would be sent without sending anything (useful on a new server):

```
sudo -u sentinelforge python3 /opt/sentinelforge/sentinelforge_linux_shipper.py --dry-run --lookback-minutes 30
```

4.4 Safe test: SSH password guessing

From another computer, try a wrong password 6 times for a real account, then sign in correctly from the same computer:

```
ssh deploy@203.0.113.10      # enter a wrong password 6 times, then the right one
```

Within 5 minutes an **SF-AUTH-001** alert appears. It shows the failures, the successful login and the source address:

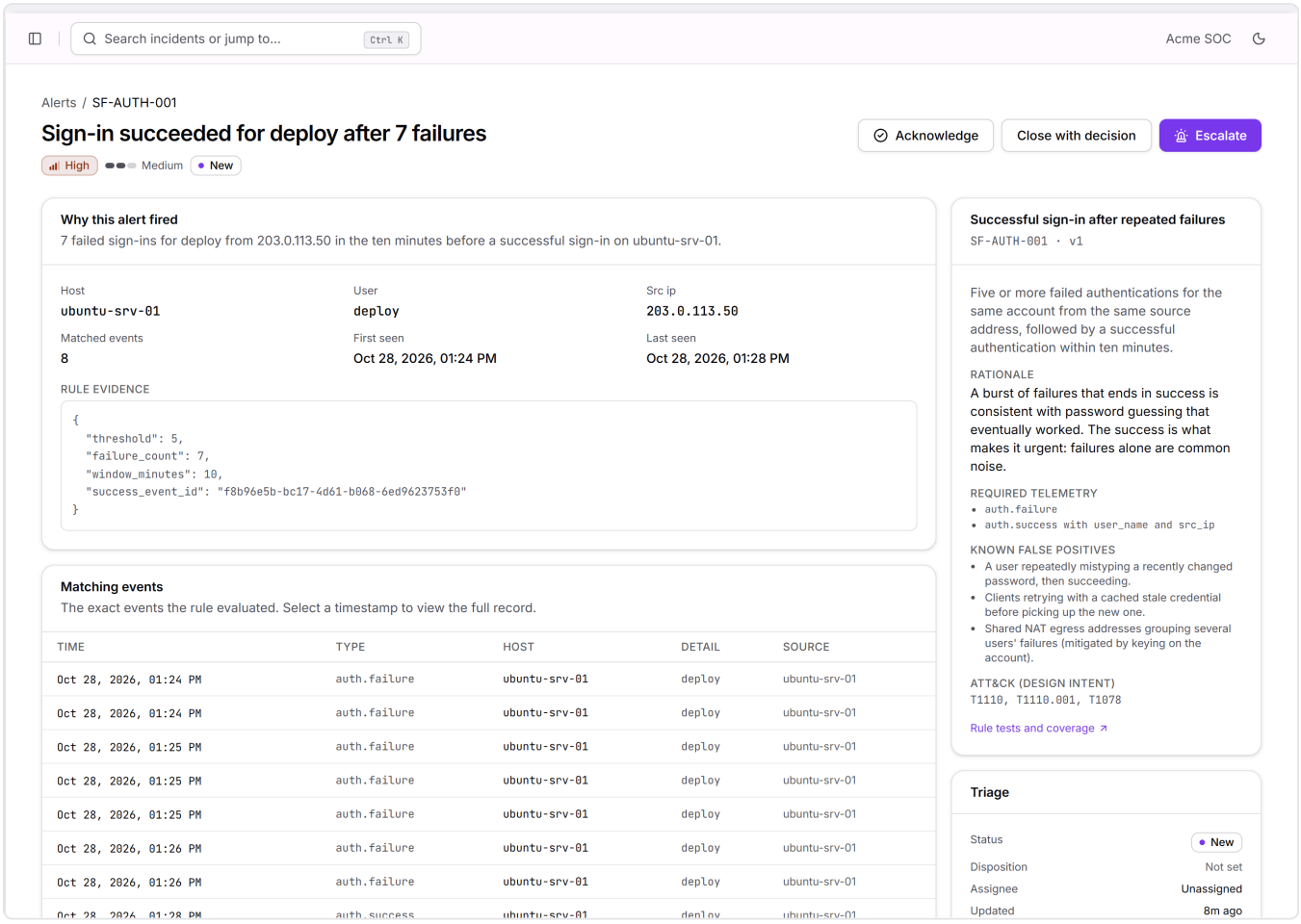


Figure 5. The alert raised by the test: 7 failed SSH logins for `deploy` from one address, then a successful one.

Password logins only

If the server allows only SSH keys (`PasswordAuthentication no` , recommended), failed attempts are still recorded but this rule cannot fire because nobody can log in with a guessed password.

4.5 What a connected server sends

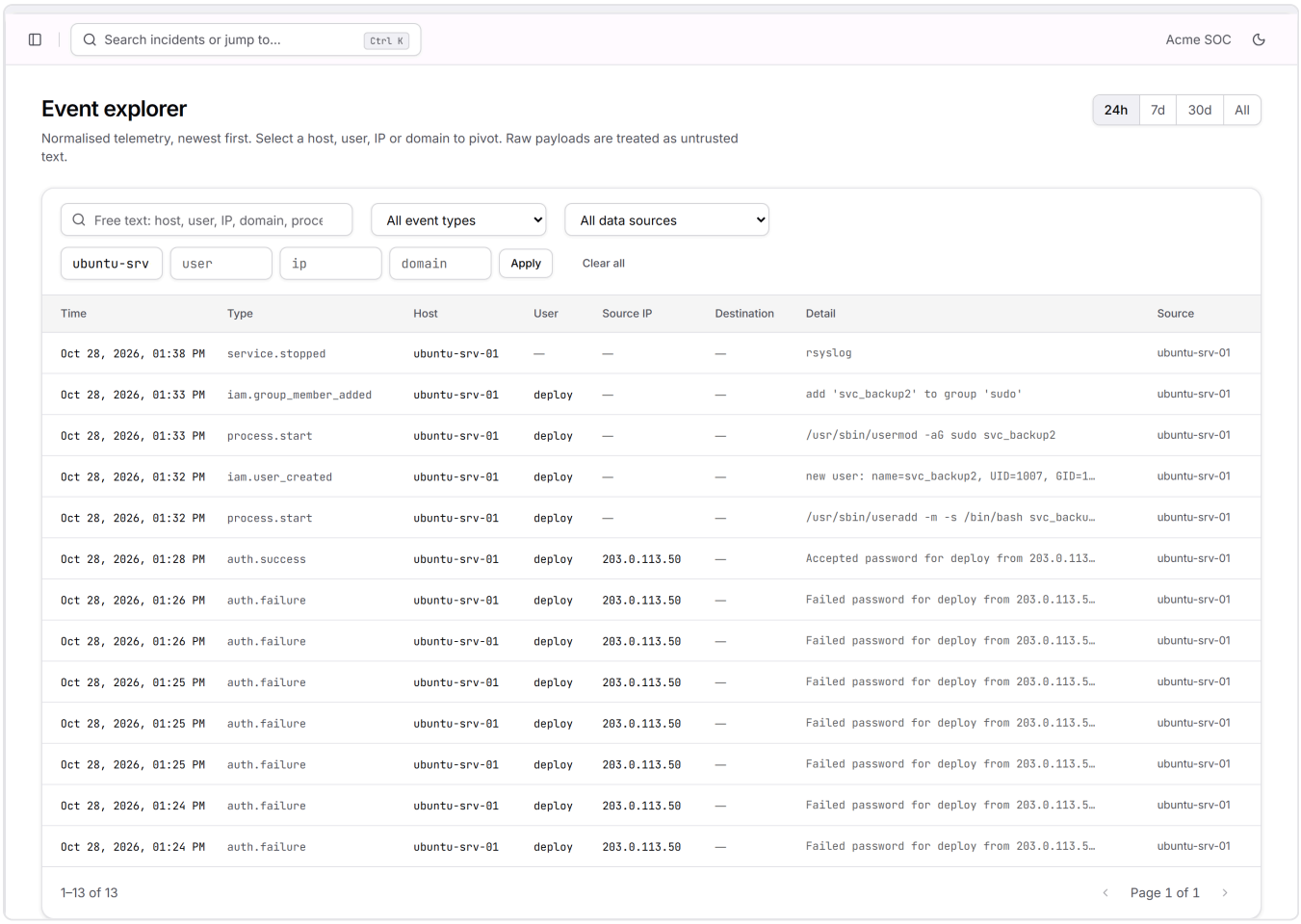


Figure 6. Events → host `ubuntu-srv-01`: SSH failures and the successful login, a new account `svc_backup2` created and added to `sudo` with the `sudo` commands that did it, and `rsyslog` stopping.

On the server	Shows up as
SSH `Failed password` / `Accepted password` / `publickey`	`auth.failure` / `auth.success` with the client address
`sudo` command	`process.start` with the full command line, parent `sudo`
`useradd`, `usermod -aG`, `gpasswd -a`, `adduser USER GROUP`	`iam.user_created`, `iam.group_member_added`; the `sudo` user who ran it is the actor
`auditd`, `rsyslog`, `journald`, Wazuh, osquery, `ufw`, AppArmor, fail2ban, ClamAV stopped	`service.stopped` (ignored during shutdown and reboot)

Not collected on Linux

Programs started without `sudo`, file access and network connections are not in the journal. They need `auditd` or an EDR agent, which this connector does not read yet.

5. Windows workstations and servers

Works on Windows 10, Windows 11 and Windows Server 2016 or later. A PowerShell script runs every 5 minutes as a scheduled task. Do these steps in **PowerShell as Administrator**: open Start, type "PowerShell", right-click **Windows PowerShell → Run as administrator**.

5.1 Turn on the auditing the detections need

On a domain, set these in a Group Policy for the machines (**Computer Configuration → Policies → Windows Settings → Security Settings → Advanced Audit Policy Configuration**). On a single machine:

```
auditpol /set /subcategory:"Logon" /success:enable /failure:enable
auditpol /set /subcategory:"Process Creation" /success:enable
auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable
auditpol /set /subcategory:"Security Group Management" /success:enable
auditpol /set /subcategory:"Other Object Access Events" /success:enable
reg add "HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\System\Audit" /v
ProcessCreationIncludeCmdLine_Enabled /t REG_DWORD /d 1 /f
```

"Other Object Access Events" records scheduled tasks being created (event 4698). The last line adds command lines to process events.

Optional but recommended: Sysmon

Install Sysmon with a community configuration (for example SwiftOnSecurity's or Olaf Hartong's). It adds LSASS access (credential dumping), Run-key writes (persistence), network connections and DNS queries. The shipper picks it up automatically and sends only LSASS access and Run-key events from those two high-volume types.

5.2 Install the shipper

1. Copy `connectors\windows\Send-WindowsEvents.ps1` to `C:\SentinelForge\` on the machine.
2. Store the ingest key (type `windows_security`, chapter 2) at machine level, so it never appears in the task's command line:

   ```
   [Environment]::SetEnvironmentVariable("SENTINELFORGE_INGEST_KEY",
   "sfk_paste_your_key_here", "Machine")
   ```

3. Open a **new** Administrator PowerShell window (so it sees the key) and run the shipper once:

   Administrator: Windows PowerShell

   ```
   PS C:\WINDOWS\system32> powershell -NoProfile -ExecutionPolicy Bypass -File
   C:\SentinelForge\Send-WindowsEvents.ps1 -ApiUrl https://sentinelforge-api.kuinkel.com
   2026-10-28 13:42:15Z sent 214 events, 214 new, 2 alerts raised
   ```

   Illustration of the expected output. The first run reads the last 60 minutes.

4 Schedule it every 5 minutes as SYSTEM:

```
$action  = New-ScheduledTaskAction -Execute "powershell.exe" -Argument '-NoProfile -
ExecutionPolicy Bypass -File "C:\SentinelForge\Send-WindowsEvents.ps1" -ApiUrl
https://sentinelforge-api.kuinkel.com'
$trigger = New-ScheduledTaskTrigger -Once -At (Get-Date) -RepetitionInterval (New-TimeSpan
-Minutes 5)
Register-ScheduledTask -TaskName "SentinelForge event shipper" -Action $action -Trigger
$trigger -User "SYSTEM" -RunLevel Highest
```

Check it in **Task Scheduler** → Task Scheduler Library → **SentinelForge event shipper**: "Last Run Result" should be **The operation completed successfully (0x0)**.

Expected one-time alert

Because this task runs PowerShell, registering it raises one **SF-PERS-001 Suspicious scheduled task** alert for the machine. Close it as **benign positive**; it is SentinelForge noticing its own installation.

5.3 What a connected Windows machine sends

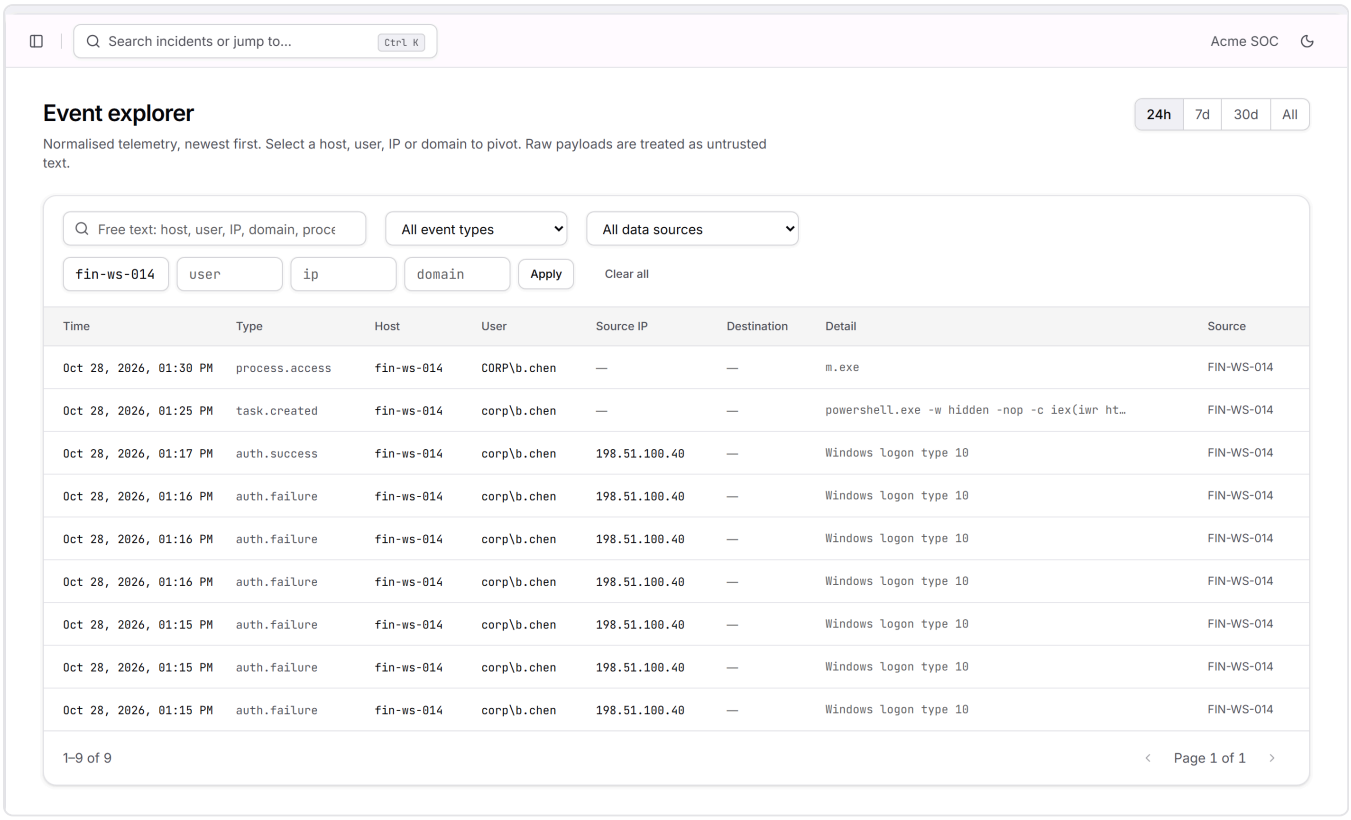


Figure 7. Events → host `fin-ws-014`: repeated RDP logon failures, a successful RDP logon from an external address, a scheduled task created, and an LSASS memory read seen by Sysmon.

Log	Event IDs	Meaning
Security	4624, 4625	Logon success / failure, with the logon type (10 = Remote Desktop)
Security	4688, 4698	Process created (with command line); scheduled task created
Security	4720, 4728, 4732, 4756	Account created; added to a global / local / universal group
Security, System	1102, 104, 7036	Log cleared; service stopped (Event Log, Sysmon, Defender)
Sysmon	1, 3, 10, 13, 22	Process, network connection, LSASS access, Run-key write, DNS query

6. Verify: events and alerts

After a machine's first run, check three places in SentinelForge.

1. **Settings → Data sources:** the machine shows **Reporting**, an event count and “last … ago” (Figure 3). A source that stays **Stale** has not delivered anything; see chapter 7.
2. **Events:** type the host name in the **host** box and click **Apply** (Figures 4, 6, 7). Click any timestamp to see the full record, including the original log fields.
3. **Alerts:** detections raised from the machines' events, newest first:

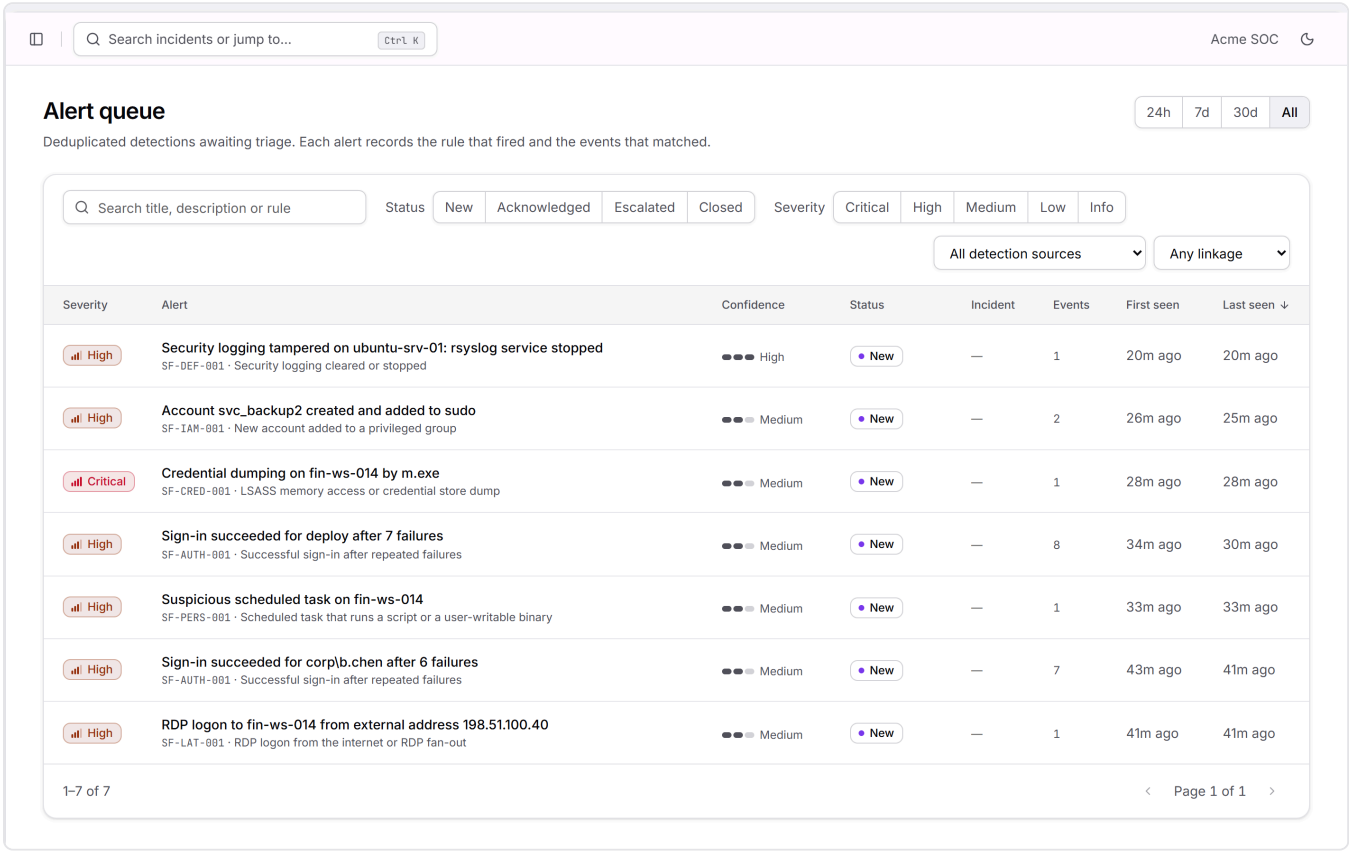


Figure 8. Alerts raised by the activity in this manual's examples: on the Ubuntu server, logging stopped, a new sudo account and SSH password guessing; on the Windows workstation, credential dumping, a suspicious scheduled task, RDP password guessing and an RDP logon from the internet.

Open an alert to see why it fired, the matching events and the rule's known false positives, then acknowledge, close with a decision, or escalate it to an incident.

7. Operations and troubleshooting

Everyday tasks

Task	Ubuntu	Windows
Run now	`sudo systemctl start sentinelforge-shipper.service`	Task Scheduler → right-click the task → Run
See the last runs	`journalctl -u sentinelforge-shipper.service -n 20`	Task Scheduler → History tab
Rotate the key	Settings → Rotate key, then `sudoedit /etc/sentinelforge/linux-shipper.env`	Settings → Rotate key, then set `SENTINELFORGE_INGEST_KEY` again (5.2 step 2)
Re-send recent history	`sudo rm /var/lib/sentinelforge/journal.cursor`	Delete `C:\ProgramData\SentinelForge\windows-events.state`
Uninstall	`sudo bash install.sh --uninstall`	`Unregister-ScheduledTask "SentinelForge event shipper"`, delete `C:\SentinelForge`, and clear the key: `[Environment]::SetEnvironmentVariable("SENTINELFORGE_INGEST_KEY", $null, "Machine")`

Re-sending is safe: SentinelForge ignores events it already has. If the API cannot be reached, the shippers keep their place and catch up on the next run.

Common problems

Symptom	Cause and fix
`HTTP 401: Missing or invalid ingest key`	Wrong, rotated or revoked key. Copy a new one (Rotate key) and update the machine.
`HTTP 429` / "Too many requests"	Rate limit per source (60 requests a minute). The shipper retries; persistent 429s mean several machines share one key: give each its own.
`API unreachable` / timeout	Outbound HTTPS is blocked. Allow TCP 443 to `sentinelforge-api.kuinkel.com`; check a proxy.
Ubuntu: `journalctl failed`	The account cannot read the journal. Re-run the installer, which adds `sentinelforge` to `systemd-journal`.
Ubuntu: runs fine but "sent 0"	No security events since the last run; normal on a quiet machine. Use `--dry-run --lookback-minutes 600` to see recent candidates.
Windows: "Set SENTINELFORGE_INGEST_KEY…"	The key was set in a window opened earlier, or for the user instead of the machine. Set it with `"Machine"` and open a new window.

Windows: no process command lines	The registry setting in 5.1 is missing; apply it and wait for new processes.
Windows: no scheduled-task or LSASS events	“Other Object Access Events” auditing is off, or Sysmon is not installed / not configured for events 10 and 13.
Source stays **Stale**	Check the last run on the machine first; any HTTP error message names the problem.

Security checklist

- ☐ One data source and key per machine.
- ☐ Keys only in the installer prompt (Ubuntu) or the machine environment variable (Windows).
- ☐ Rotate a machine's key when it is rebuilt, sold or given to someone else; revoke it when the machine is retired.
- ☐ Close the shipper's own SF-PERS-001 alert on Windows as benign positive, once per machine.
- ☐ Review new alerts daily; each one links to the events that caused it.